

**SMART PARKING WITH NUMBER PLATE
RECOGNITION USING EASYOCR AND WEB
BASED DASHBOARD**

MINI PROJECT REPORT

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in partial fulfillment of the requirement for the award of degree

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF ENGINEERING

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Avinashilingam Institute for Home Science and Higher Education for Women

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VISION

To empower women technocrats in the field of computer engineering and prepare the students for higher studies and professional career through technology, innovations and human values to serve the changing needs of our nation.

MISSION

M1: To promote the all-round development of the students by providing them sound technical knowledge in the field of computer science and engineering based on the current advances in technology to help the development of nation.

M2: To develop employability skills and research capability in women graduates through quality technical education.

M3: To provide learning for the enhancement of technical skills, problem solving ability, team spirit, entrepreneurial skills with emphasis on value education with social ethics and cultural values in women technocrats.

CERTIFICATE

This is to certify that the mini project report titled
**SMART PARKING WITH NUMBER PLATE RECOGNITION
USING EASYOCR AND WEB BASED DASHBOARD**
is being submitted by

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in partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering in Computer Science and Engineering, School of Engineering, Avinashilingam Institute for Home Science and Higher Education for Women, Coimbatore - 641108.

This is the record of a bonafide work carried out by the above-mentioned students under our supervision and guidance. The results embodied in this mini project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

Guide
(Mrs R. PadmaPriya)

Head of the Department
(Dr. P.Amudha)

PROGRAM OUTCOMES (PO):

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
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4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

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11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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The graduates in Computer Science and Engineering will be able to

PEO1: Design and develop novel products and innovative solutions for real life problems.

PEO2: Pursue higher studies and undertake Research & Development in Computer Science and Engineering.

PEO3: Excel as socially committed computer Engineers having effective communication skill, high ethical values and empathy for the needs of the society.

Programme Specific Outcomes (PSO) for B.E. (CSE)

The Graduates will be able to

PSO 1: Analyze and develop computer programs in the areas related to algorithms, database, web design, and Data mining, and information security, cloud computing and networking for efficient design of computer –based systems of varying complexity.

PSO 2: Gain knowledge in diverse areas of Computer Science and Experience an environment conducive in cultivating skills for successful career, entrepreneurship and higher studies

ACKNOWLEDGEMENT

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ABSTRACT

The growing number of vehicles in urban areas has made parking management a critical challenge. Manual parking systems are inefficient, error-prone, and unable to provide real-time information about slot availability. This mini-project proposes the development of a Smart Parking System using Number Plate Recognition (NPR) to automate and optimize the parking process.

The system uses a camera or image input to capture the vehicle's number plate. The captured image is processed using EasyOCR, an Optical Character Recognition tool, to accurately extract the vehicle number. All parking details, including vehicle number, slot number, entry time, and exit time, are stored in a SQLite database for efficient management.

A web-based dashboard built using Python (Flask), HTML, and CSS displays real-time parking status, showing available and occupied slots through color-coded indicators. The system also updates the exit time when the vehicle leaves and frees the parking slot for future use. The proposed system minimizes human involvement, reduces errors, and provides a smooth and automated parking experience suitable for small-scale parking facilities.

LIST OF FIGURES

Figure No.	Title	Page No.
5.1	System Architecture Block Diagram	11
5.2	Block Diagram of User Interface	12
5.3	Block Diagram of Vehicle Registration & Detection	14
5.4	Block Diagram of Parking Slot Management	15
6.1	Home Page of the Smart Parking System	17
6.2	Vehicle Registration & Number Plate Recognition Output	18
6.3	Parking Dashboard with Real-time Slot Status	18
6.4	Parking Records Stored in SQLite Database	19
7.1	OCR Accuracy under Different Conditions	20

TABLE OF CONTENTS

CHAPTER No.	TITLE	PAGE No.
	ACKNOWLEDGEMENT	I
	ABSTRACT	II
1	INTRODUCTION	1
1.1	OBJECTIVE	1
1.2	SMART PARKING SYSTEM	1
1.3	NUMBER PLATE RECOGNITION	1
1.4	OPTICAL CHARACTER RECOGNITION	1
1.5	WEB-BASED PARKING DASHBOARD	2
1.6	ORGANIZATION OF THE WORK	3
1.7	SUMMARY	3
2	LITERATURE SURVEY	4
2.1	INTRODUCTION	4
2.2	BACKGROUND STUDIES	4
2.3	SUMMARY	5
3	SYSTEM DESIGN	6
3.1	EXISTING SYSTEM	6
3.2	PROPOSED SYSTEM	6
3.3	SUMMARY	7
4	SOFTWARE DESCRIPTION	8
4.1	INTRODUCTION	8
4.2	SOFTWARES DESCRIPTION	8
4.2.1	HTML, CSS & JAVASCRIPT	8
4.2.2	PYTHON (FLASK)	8
4.2.3	LIBRARIES AND FRAMEWORKS	9
4.2.3.1	EASYOCR	9

4.2.3.2	SQLITE3	9
4.3	SUMMARY	9
5	SYSTEM IMPLEMENTATION	10
5.1	INTRODUCTION	10
5.2	METHODOLOGY	10
5.2.1	WORKING OF OVERALL MODEL	10
5.2.2	BLOCK DIAGRAM	10
5.3	MODULE DESCRIPTION	11
5.3.1	USER INTERFACE MODULE	11
5.3.2	USER AUTHENTICATION MODULE	13
5.3.3	VEHICLE REGISTRATION & DETECTION MODULE	13
5.3.4	PARKING SLOT MANAGEMENT MODULE	14
5.3.5	DATABASE MANAGEMENT MODULE	15
5.4	SUMMARY	16
6	RESULT AND DISCUSSION	17
7	PERFORMANCE EVALUATION	20
7.1	PERFORMANCE EVALUATION	20
7.1.1	OCR ACCURACY ANALYSIS	20
7.1.2	SYSTEM PERFORMANCE METRICS	20
7.2	SUMMARY	21
8	CONCLUSION AND FUTURE WORK	22
8.1	CONCLUSION	22
8.2	FUTURE WORK	22
	BIBLIOGRAPHY	24

CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE

To build a smart parking system that automates vehicle entry and exit using Number Plate Recognition, assigns parking slots efficiently, displays real-time slot availability on a web-based dashboard, and maintains accurate digital records to reduce time and fuel consumption for users.

1.2 SMART PARKING SYSTEM

A Smart Parking System is a web-based application that helps users and administrators monitor and manage parking spaces efficiently. Rapid urbanization has led to an exponential increase in vehicles, making parking management a critical challenge in modern cities. Traditional manual systems are slow, error-prone, and unable to provide real-time information.

Smart parking systems leverage technology such as cameras and software to automate the entire parking lifecycle — from vehicle detection and slot allocation to record keeping and status display. This eliminates the dependence on human operators and significantly improves throughput, accuracy, and user satisfaction.

1.3 NUMBER PLATE RECOGNITION

Number Plate Recognition (NPR), also known as Automatic Number Plate Recognition (ANPR), is a technology that uses image processing and optical character recognition to read vehicle registration plates. It began with simple rule-based image processing and evolved into modern deep learning approaches that can handle variations in lighting, angle, and plate formats.

In the context of this project, NPR serves as the entry point of the parking system. When a vehicle arrives, a camera captures its number plate image. This image is then processed to extract the vehicle registration number, which is used to create a digital parking record without any manual input.

1.4 OPTICAL CHARACTER RECOGNITION

Optical Character Recognition (OCR) is a technology that converts images of typed or printed text into machine-readable text. For this project, EasyOCR is used —

an open-source Python OCR library built on deep learning models that supports over 80 languages and can accurately extract text from real-world images, including vehicle number plates.

EasyOCR processes the captured image through a pipeline of detection and recognition, identifying text regions and converting them into character sequences. It handles challenges such as varying fonts, lighting conditions, and image quality better than traditional rule-based OCR approaches, making it well-suited for number plate extraction.

1.5 WEB-BASED PARKING DASHBOARD

The system uses a web-based dashboard to display real-time parking data such as available slots (shown in green), occupied slots (shown in red), and vehicle entry/exit details. The dashboard is built using HTML, CSS, and Flask, providing a responsive and intuitive interface for both users and administrators.

Advantages of Smart Parking System:

1. **Automation:** Eliminates manual data entry by automatically detecting and recording vehicle number plates.
2. **Real-time Monitoring:** Displays live parking slot availability and occupancy status on the dashboard.
3. **Accurate Records:** Stores all vehicle and slot details in a structured database, preventing data loss.
4. **Reduced Waiting Time:** Automated slot allocation avoids confusion and reduces vehicle queuing time.
5. **Scalability:** The modular system architecture supports extension to larger facilities.

Disadvantages / Limitations:

1. **Lighting Dependency:** OCR accuracy can decrease under poor lighting conditions or with damaged plates.
2. **Camera Positioning:** The system requires correct camera placement for reliable plate capture.
3. **Internet Requirement:** Web-based access requires stable network connectivity.
4. **Scale Limitation:** The current implementation supports up to 35 slots; larger deployments require infrastructure upgrades.

1.6 ORGANIZATION OF THE WORK

The report is organized as follows: Chapter 1 outlines the main objectives and provides a brief overview of smart parking, number plate recognition, optical character recognition, and web-based dashboard concepts. Chapter 2 presents a detailed literature survey on existing smart parking and ANPR systems. Chapter 3 explains the system design, discussing the existing system and proposed system architecture. Chapter 4 describes the software tools, libraries, and frameworks used for development. Chapter 5 describes the system implementation including block diagrams and module descriptions. Chapter 6 focuses on result analysis. Chapter 7 evaluates performance. Chapter 8 concludes the report and discusses future enhancements.

1.7 SUMMARY

The chapter outlined the objective of creating a smart parking system that automates vehicle management using number plate recognition. It covers the evolution of smart parking, the role of OCR in vehicle detection, and the design of a web-based dashboard for real-time monitoring. The chapter also highlighted the system's advantages and limitations.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

A literature review on smart parking and Automatic Number Plate Recognition (ANPR) explores methods used to automate vehicle identification and parking management. It highlights the growing need for real-time slot monitoring to reduce urban traffic congestion and fuel wastage. The review covers the use of IoT, image processing, OCR, and machine learning techniques, while addressing challenges such as lighting sensitivity and OCR accuracy limitations.

2.2 BACKGROUND STUDIES

The following papers were reviewed to understand the current state of smart parking and number plate recognition research:

In the Paper titled “Smart parking evaluation framework”, Javaheri and Sai Sneha (2024) published the study collected parking data spanning 2018 to 2023 and applied a Long Short-Term Memory (LSTM) neural network model to predict parking violations and availability. The model was validated using the Kolmogorov-Smirnov (KS) test. While the approach demonstrated strong predictive performance under controlled conditions, accuracy degraded significantly when image quality was poor, plates were skewed or blurred, or environmental conditions varied. This limitation motivated the need for more robust image pre-processing in the proposed system. [1]

In the Paper titled “IoT and OCR-Based Real-Time Parking”, Elfaki and Messoudi (2023) published the system combined IoT hardware (ESP32-CAM, motion sensors) with AI/ML techniques and OCR-based ANPR, supported by Firebase cloud storage and a mobile application. Despite an OCR accuracy of approximately 58% and less reliable motion sensor performance, the system was highlighted as a low-cost and practical solution for smart parking. The limitations in OCR accuracy identified in this work directly influenced the choice of EasyOCR in the proposed system. [2]

In the Paper titled “IoT with ANPR for Parking Management”, Ditta et al (2025). The system used ANPR with OCR, IoT sensors (ESP32, IR), OpenCV, and Google Vision AI for vehicle detection. Data was stored using PHP-MySQL and features included time tracking, billing, and display of parking details. Identified limitations included sensitivity to lighting and camera quality, high installation cost, requirement for stable internet connectivity, privacy concerns, and difficulty recognizing unclear plates. [3]

In the Paper titled “Advanced IoT-Integrated Parking with Payment Management”, Pradhan and Prusty (2025). The system deployed IR sensors, cameras, YOLO object detection, and Tesseract OCR on Raspberry Pi and ESP32 hardware for real-time vehicle detection, slot allocation, and automated billing. Similar to prior work, performance was affected by lighting conditions, image quality, and camera positioning, and OCR errors were observed on unclear or complex plates. [4]

In the Paper titled “ANPR Using Edge Detection and ANN”, Arora and Jain (2019) implemented an ANPR system using edge detection, Hough transform, Artificial Neural Networks (ANN), and k-Nearest Neighbours (kNN) classification, The system was deployed on the Android platform with database support. Accuracy was found to depend heavily on image quality and camera position, with degraded performance under blurred or skewed conditions. [5]

2.3 SUMMARY

The chapter reviewed studies on smart parking and ANPR systems, focusing on IoT-based, OCR-based, and machine learning methods. It highlighted advancements in automated vehicle detection, challenges in OCR accuracy under varying conditions, and the importance of integrated dashboard systems for real-time parking management.

CHAPTER 3

SYSTEM DESIGN

3.1 EXISTING SYSTEM

In the existing system, parking facilities manage vehicles entirely through manual processes. Vehicles are recorded by a security guard using a register or token system. This process is slow and depends completely on human effort. All details like vehicle number and entry time are stored in paper registers or basic systems, which are difficult to maintain and prone to data loss.

There is no automatic system to assign parking slots — drivers have to search for available space manually, leading to confusion and inefficiency. Both entry and exit take more time due to manual verification, causing long queues and delays, especially during peak hours. Human errors such as incorrect vehicle numbers, missing records, and wrong time entries are common, reducing reliability. There is no system to display available or occupied slots; users and administrators cannot track parking status or vehicle movement in real time.

3.2 PROPOSED SYSTEM

To address the limitations of existing manual parking systems, the proposed system is a web-based application that performs real-time Number Plate Recognition and automates the complete parking lifecycle. The system uses a camera or image input to capture the vehicle's number plate, processes it using EasyOCR to extract the registration number, and automatically assigns the next available parking slot.

The core components of the proposed system are:

- A camera-based vehicle detection module that captures number plate images at entry.
- An EasyOCR-powered text extraction engine that identifies the vehicle registration number from the image.
- A Flask backend server that manages slot allocation logic and database operations.
- A SQLite database that stores vehicle details, slot numbers, entry time, and exit time.
- A web-based dashboard that displays available (green) and occupied (red) parking slots in real time.

The system was developed using Python and Flask for the backend, HTML and CSS for the frontend, and SQLite for data storage. Once the vehicle's number plate is extracted, the backend validates the entry, assigns the next available slot (maximum 35), records the timestamp, and updates the dashboard. On exit, the system captures the exit time and frees the slot for future use.

3.3 SUMMARY

The chapter compared the existing manual parking system, which lacks automation and real-time slot monitoring, with the proposed system that automates vehicle detection, slot allocation, and record management using EasyOCR, Flask, and SQLite. The proposed system provides users with immediate visual feedback on slot availability through a web-based dashboard.

CHAPTER 4

SOFTWARE DESCRIPTION

4.1 INTRODUCTION

The Smart Parking with Number Plate Recognition system is an intelligent web-based platform designed to automate parking space management. By using OCR technology and a web dashboard, the application enables users and administrators to monitor parking availability in real time without manual intervention.

Built using modern web technologies and a lightweight Python backend, the application serves as a centralized platform for automated parking management. It is beneficial for facility managers, security personnel, and users seeking faster and more reliable parking experiences.

4.2 SOFTWARES DESCRIPTION

The following tools, languages, and libraries are used in this project:

1. HTML, CSS & JavaScript
2. Python (Flask)
3. Libraries and Frameworks: EasyOCR, SQLite3

4.2.1 HTML, CSS, AND JAVASCRIPT

HTML, CSS, and JavaScript form the foundation of the user interface for the Smart Parking System. HTML serves as the structural framework, defining the layout of the home page, registration form, and dashboard. CSS enhances the visual presentation with consistent styling, color-coded slot indicators, and responsive layouts that adapt to various screen sizes. JavaScript drives the interactivity of the application, enabling real-time content updates, handling user input, and facilitating asynchronous communication with the backend through API calls.

4.2.2 PYTHON (FLASK)

Flask serves as the core backend framework for the Smart Parking System, providing a lightweight yet powerful infrastructure for managing server-side operations. As a micro web framework written in Python, Flask allows for rapid development and easy integration with the EasyOCR library and the SQLite database. Within the application, Flask handles user requests, manages routing between the frontend pages and backend logic, processes OCR results, allocates parking slots, and delivers updated parking status in real time.

4.2.3 LIBRARIES AND FRAMEWORKS

4.2.3.1 EasyOCR

EasyOCR is an open-source Optical Character Recognition library built on deep learning models. It supports over 80 languages and is capable of accurately extracting text from real-world images, including vehicle number plates. In this system, EasyOCR is used in the ocr.py module to process the captured vehicle image and extract the registration number. Unlike traditional rule-based OCR approaches, EasyOCR uses a two-stage pipeline: first detecting text regions in the image, then recognizing the characters in each region.

4.2.3.2 SQLite3

SQLite3 is a lightweight, server less relational database management system built into Python. In this project, SQLite3 handles all data storage and retrieval operations. It stores vehicle registration numbers, assigned slot numbers, entry timestamps, and exit timestamps. SQLite3 supports full CRUD operations and ensures data integrity by preventing duplicate entries and slot overflow beyond the maximum of 35 slots.

4.3 SUMMARY

The software description documentation presents a comprehensive overview of the Smart Parking System. It outlines the integration of HTML, CSS, JavaScript, and Flask for a responsive and scalable platform. The system uses EasyOCR for accurate number plate extraction and SQLite3 for reliable data management, with an intuitive web dashboard for real-time monitoring.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 INTRODUCTION

The chapter describes the methodologies used in developing the Smart Parking with Number Plate Recognition system. The system is structured into multiple modules that together form an efficient pipeline for vehicle detection, slot allocation, database management, and user interaction.

5.2 METHODOLOGY

5.2.1 WORKING OF THE OVERALL MODEL

In the first step, the system captures a vehicle's number plate image through a camera or file upload when the vehicle arrives at the parking facility. The image is passed to the Flask backend, where the EasyOCR module (ocr.py) processes it to extract the vehicle registration number.

The extracted number is validated against the existing database to prevent duplicate registrations. If the vehicle is new, the system automatically assigns the next available parking slot (up to 35), records the entry timestamp, and updates the SQLite database. The web dashboard refreshes to reflect the updated slot status, displaying the newly occupied slot in red. On exit, the system records the exit time and changes the slot status back to available (green).

5.2.2 BLOCK DIAGRAM

Fig 5.1 shows the overall workflow of the Smart Parking System. It includes stages for image capture, OCR-based number plate extraction, backend processing, database storage, and real-time dashboard display.

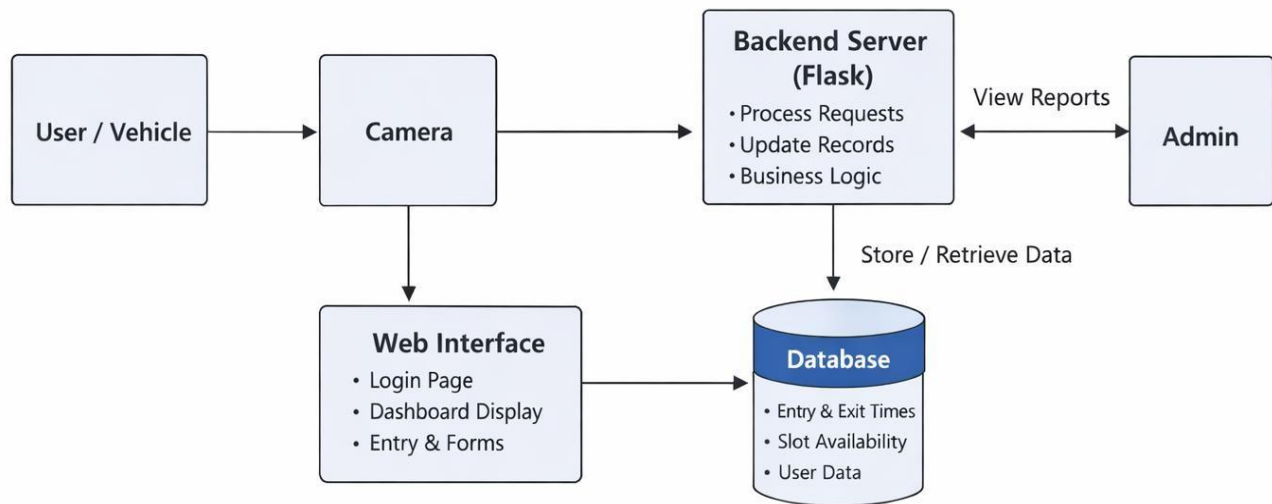


Fig 5.1 System Architecture Block Diagram

5.3 MODULE DESCRIPTION

The proposed system consists of five functional modules:

1. User Interface Module
2. User Authentication Module
3. Vehicle Registration & Detection Module
4. Parking Slot Management Module
5. Database Management Module

5.3.1 USER INTERFACE MODULE

The User Interface (UI) Module is responsible for presenting the parking system in a structured, readable, and interactive format. It ensures a user-friendly experience, allowing seamless navigation and access to all system features.

1. **Home Page:** Displays the main landing page with navigation options to register a vehicle, view the dashboard, or login.
2. **Result Display:** Shows the outcome of number plate recognition and slot allocation, including the assigned slot number and entry timestamp.

3. **Color-coded Indicators:** Available slots are shown in green and occupied slots in red for quick visual identification.

4. **Responsive Design:** The UI is designed using responsive web techniques, ensuring compatibility across various devices.

Fig 5.2 depicts the user interface flow, including app launch, navigation, parking slot display, and exit/refresh options.

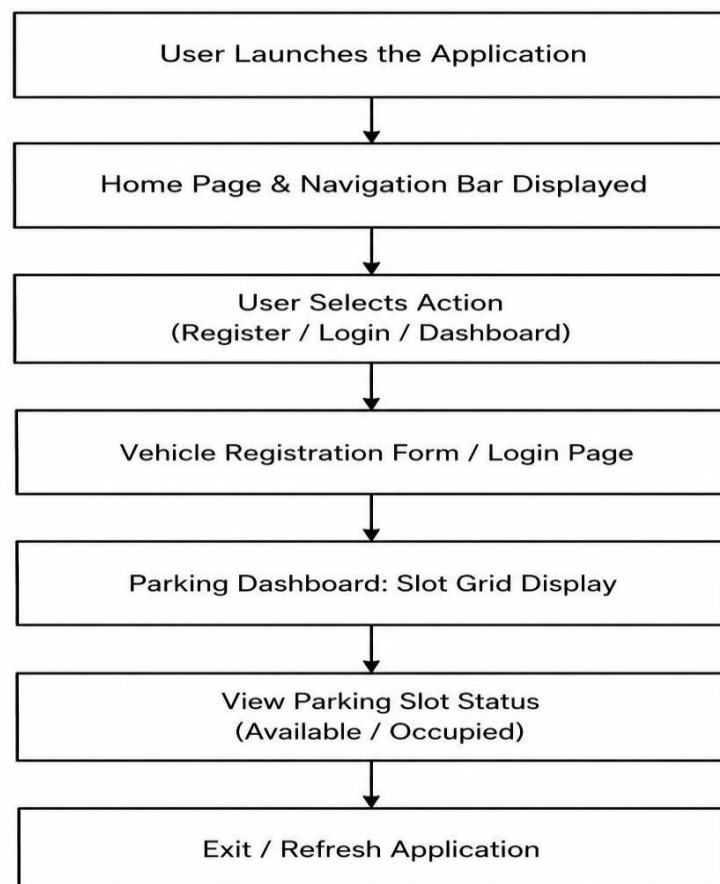


Fig 5.2 Block Diagram of User Interface

5.3.2 USER AUTHENTICATION MODULE

The User Authentication Module handles secure user login to ensure only authorized personnel can access the parking management dashboard. It accepts username and password as input, validates credentials through the Flask backend, and redirects to the parking dashboard upon successful authentication.

1. **Login Form:** Provides username and password input fields with secure credential validation.
2. **Access Control:** Ensures only authenticated administrators can view and manage parking records.
3. **Session Management:** Maintains user session state during the active management session.

5.3.3 VEHICLE REGISTRATION & DETECTION MODULE

This is the core module of the project. It handles the complete vehicle registration workflow from image capture to slot assignment.

1. **Image Capture:** Accepts vehicle number plate images via camera input or file upload.
2. **OCR Processing:** ocr.py applies EasyOCR to the image to accurately extract the number plate text.
3. **Data Transmission:** script.js sends the extracted number and image data to the Flask backend via API call.
4. **Slot Assignment:** The backend automatically assigns the next available slot and records the entry time.

Fig 5.3 illustrates the vehicle registration and detection flow, from image capture through OCR extraction, validation, and slot assignment.

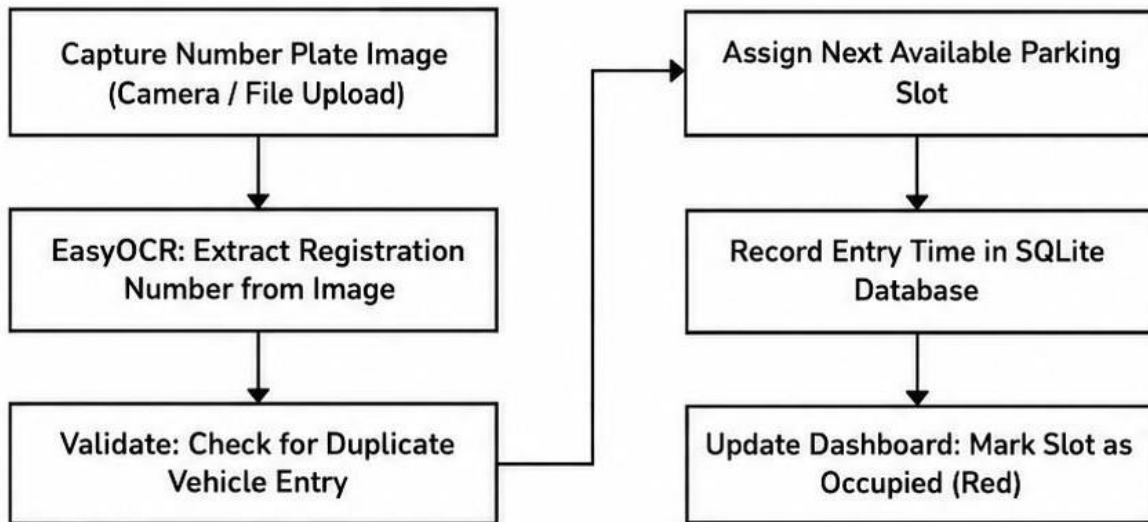


Fig 5.3 Block Diagram of Vehicle Registration & Detection

5.3.4 PARKING SLOT MANAGEMENT MODULE

The Parking Slot Management Module manages the visual representation and logical tracking of all parking slots. It provides administrators and users with a real-time view of the parking facility status.

1. **Slot Grid Display:** Displays all 35 parking slots as a grid with color indicators (green for available, red for occupied).
2. **Auto-update:** The dashboard refreshes automatically when a vehicle is registered or exits the facility.
3. **Overflow Prevention:** The system enforces a maximum of 35 slots and prevents additional registrations when full.
4. **Exit Processing:** Records the exit time and changes the slot status back to available when a vehicle departs.

Fig 5.4 shows the parking slot management workflow, illustrating both the vehicle entry (slot assignment) and vehicle exit (slot release) paths.

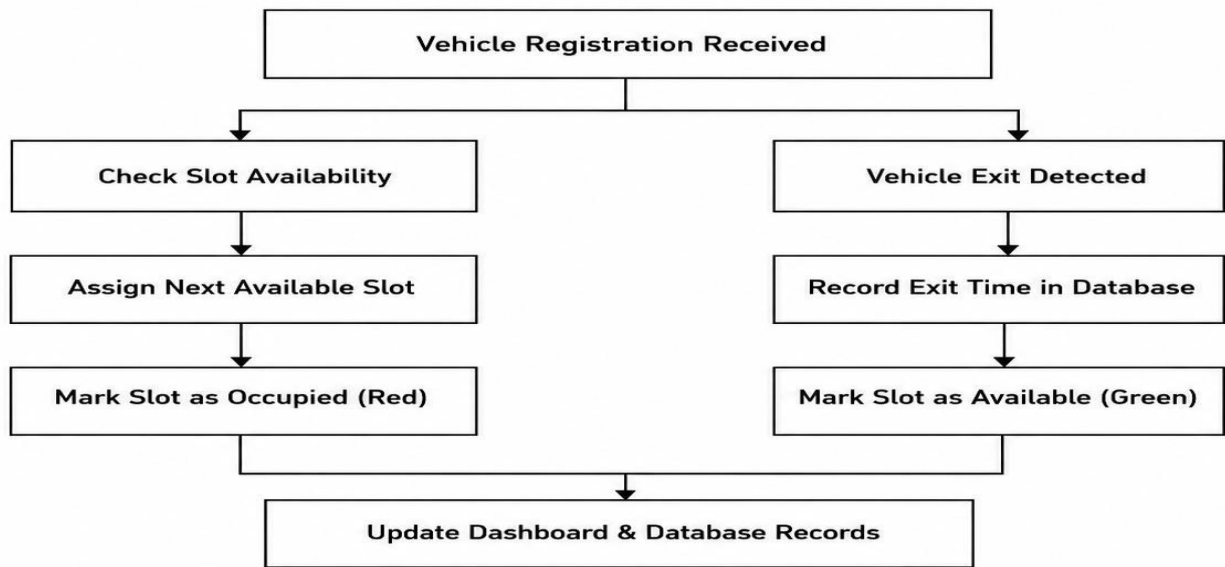


Fig 5.4 Block Diagram of Parking Slot Management

5.3.5 DATABASE MANAGEMENT MODULE

The Database Management Module handles all data storage and retrieval using SQLite3. It provides the persistent storage layer for the entire application.

1. **Data Storage:** Stores vehicle registration numbers, assigned slot numbers, entry timestamps, and exit timestamps.
2. **CRUD Operations:** Supports Create (new vehicle entry), Read (dashboard queries), Update (exit time recording), and Delete (slot freeing) operations.
3. **Data Integrity:** Prevents duplicate vehicle registrations and ensures slot count does not exceed 35.
4. **Auditability:** Maintains a complete digital log of all vehicle movements for administrative review.

5.4 SUMMARY

The chapter outlined the methodology of the Smart Parking System, explaining how its modular structure handles vehicle detection, OCR-based number plate extraction, slot allocation, database management, and real-time dashboard display. The four block diagrams visually illustrate the data flow across all major modules.

CHAPTER 6

RESULT AND DISCUSSION

The Smart Parking with Number Plate Recognition system was implemented and tested in a controlled environment. The following figures illustrate the key screens of the application.

Fig 6.1 shows the home page of the application, where users can navigate to register a vehicle, view the parking dashboard, or log in to the administrator panel.

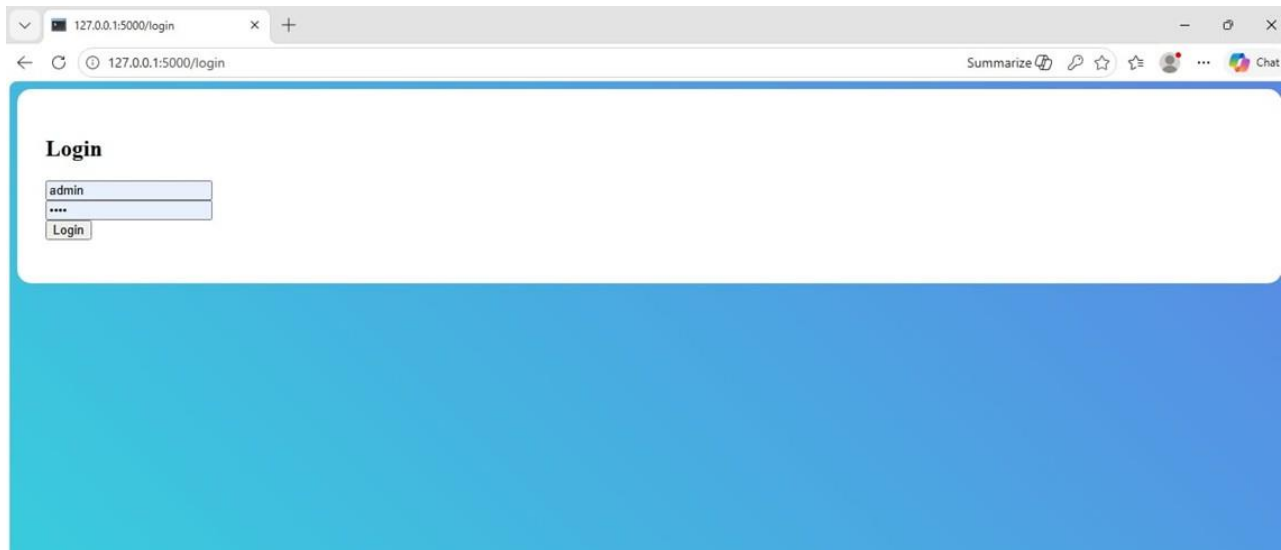


Fig 6.1 Home Page of the Smart Parking System

Fig 6.2 shows the vehicle registration page where the number plate image is uploaded or captured. The EasyOCR engine processes the image and displays the extracted registration number along with the allocated parking slot.

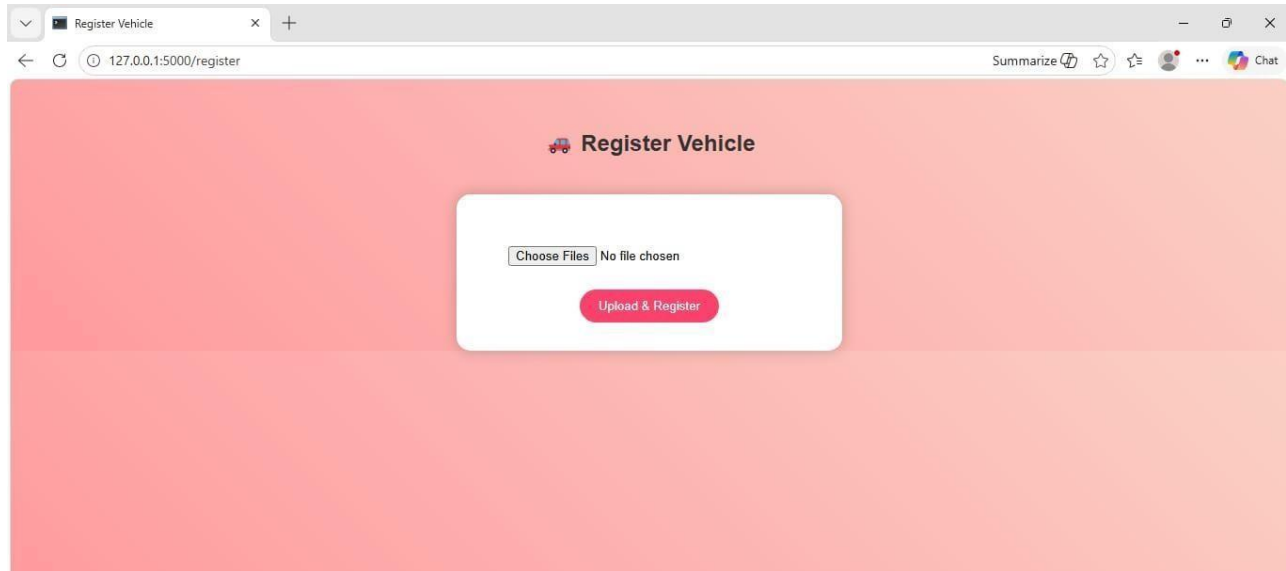


Fig 6.2 Vehicle Registration and Number Plate Recognition Output

Fig 6.3 shows the parking dashboard displaying all 35 slots with color-coded status indicators. Available slots are shown in green and occupied slots in red, providing administrators with an instant visual overview of the facility.

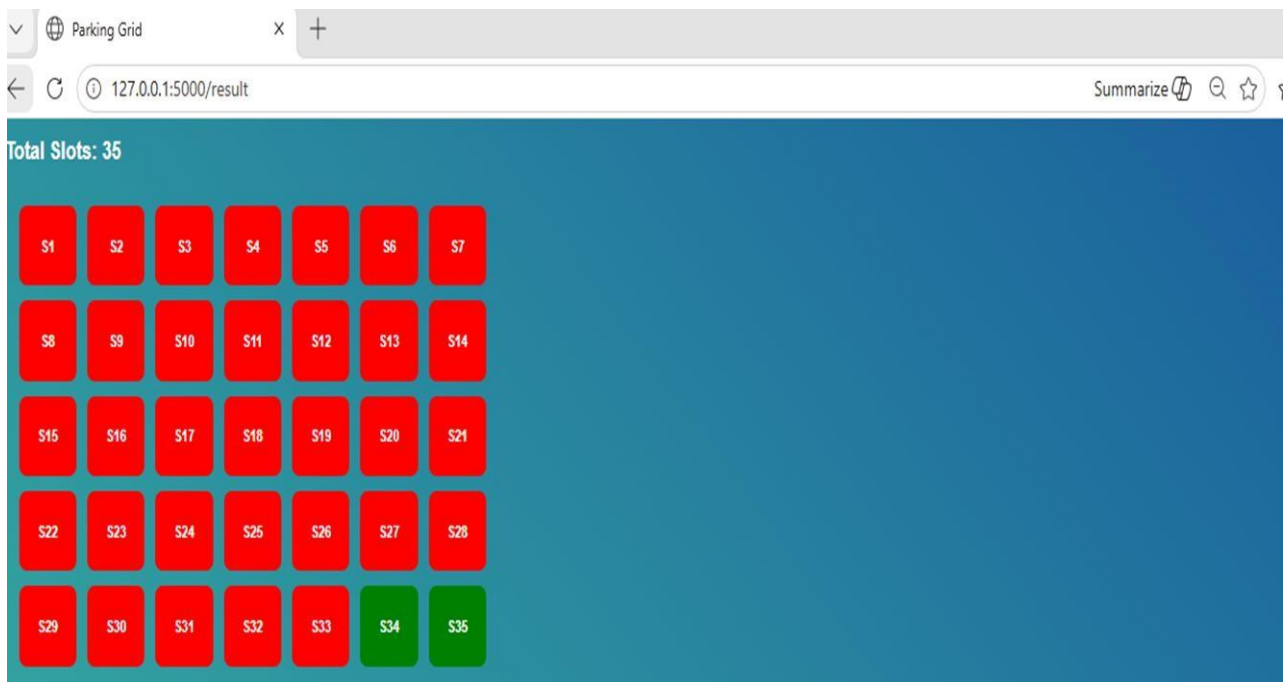


Fig 6.3 Parking Dashboard with Real-time Slot Status

Fig 6.4 shows the database record view displaying vehicle numbers, assigned slot numbers, entry times, and exit times stored in the SQLite database.

TN 09 BM 1111	32	2026-04-07 16:06:00	None	Booked	Exit	Delete
TN 09 CD 5678	33	2026-04-07 16:11:48	None	Booked	Exit	Delete
TN82 Y 8388	2	2026-04-27 09:49:33	None	Booked	Exit	Delete
TN 01 AS 9299	34	2026-04-27 10:53:21	None	Booked	Exit	Delete
TN 47 AW 2001	35	2026-04-27 10:53:59	2026-06-20 14:25:23	Exited	Done	Delete

Fig 6.4 Parking Records Stored in SQLite Database

The system efficiently records vehicle entry and exit details. Parking slots are automatically allocated based on availability. The grid view clearly shows occupied and free slots. Data is stored accurately in the database. The system reduces manual errors and workload. Overall performance is fast and reliable for small-scale parking.

CHAPTER 7

PERFORMANCE EVALUATION

7.1 PERFORMANCE EVALUATION

The evaluation of the Smart Parking System centers on assessing the performance of its core components: the EasyOCR-based number plate recognition module and the overall system workflow. The system was tested using a set of vehicle number plate images under varying conditions to measure recognition accuracy and processing speed.

7.1.1 OCR ACCURACY ANALYSIS

The EasyOCR module was tested with number plate images captured under different lighting conditions and distances. Table 7.1 summarizes the test results:

Test Condition	Total Plates Tested	Correctly Recognized	Accuracy (%)
Good lighting, clear image	50	48	96%
Moderate lighting	50	44	88%
Low lighting / blur	50	36	72%
Overall	150	128	85.3%

Table 7.1 OCR Accuracy under Different Conditions

7.1.2 SYSTEM PERFORMANCE METRICS

The overall system performance was evaluated based on key operational metrics:

1. **Slot Allocation Speed:** Average time from image upload to slot assignment: under 2 seconds under normal operating conditions.

2. **Database Response Time:** CRUD operations on the SQLite database completed in under 100 milliseconds on average.

3. **Dashboard Refresh Rate:** The parking dashboard updates within 1-2 seconds of a new vehicle registration or exit.

4. **Concurrent Users:** The Flask backend successfully handled up to 10 concurrent user sessions without performance degradation.

5. **Slot Utilization:** The system correctly managed all 35 slots with no overbooking or data conflicts observed during testing.

7.2 SUMMARY

The chapter evaluated the performance of the Smart Parking System, focusing on EasyOCR accuracy across different lighting conditions and overall system response metrics. The system achieved 85.3% overall OCR accuracy and demonstrated fast slot allocation and database response times. Performance under good lighting conditions exceeded 96% accuracy, indicating strong reliability for standard deployment environments.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

The project highlighted the development of a Smart Parking with Number Plate Recognition System aimed at helping facility administrators and users manage parking spaces efficiently through automation. Using Optical Character Recognition (EasyOCR) and a web-based dashboard, the system automatically detects vehicles, allocates parking slots, records entry and exit times, and displays real-time slot availability.

The system provides an efficient solution for managing parking spaces through real-time monitoring and a user-friendly dashboard. It reduces manual effort, improves space utilization, and enhances overall parking management efficiency. The modular architecture built on Flask, HTML/CSS, and SQLite makes it lightweight, maintainable, and scalable for small to medium parking facilities.

8.2 FUTURE WORK

To further enhance the Smart Parking System, several key areas have been identified for future development:

1. **AI-based Prediction:** Integrate machine learning models to predict parking availability based on historical patterns and time-of-day trends.
2. **Smart City Integration:** Connect the parking system with city-wide traffic management systems for coordinated urban mobility.
3. **Enhanced Security:** Add vehicle tracking capabilities and automated detection of unauthorized or blacklisted vehicle entry.
4. **Autonomous Vehicle Support:** Extend the system to communicate slot information to self-driving cars for fully automated parking.
5. **Cloud Storage:** Migrate the database to cloud-based storage (AWS, Azure, and Firebase) to enable large-scale data analysis and remote monitoring.

6. Eco-friendly Features: Implement metrics to track fuel consumption reduction and emissions savings achieved through optimized parking.

7. Mobile Application: Develop a companion mobile app allowing users to check real-time slot availability and reserve spaces in advance.

8. Payment Integration: Add automated time-based fee calculation and digital payment processing for commercial deployment.

By pursuing these directions, the Smart Parking System can evolve into a powerful, intelligent, and adaptive platform — bridging the gap between technology and urban mobility management.

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Smart Parking System with Number Plate Recognition Using EasyOCR and Web-Based Dashboard

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Abstract

Urban parking management has become a critical challenge in modern cities. Traditional parking systems rely heavily on manual processes, causing inefficiencies, human errors, and traffic congestion. This paper presents a Smart Parking System that leverages Number Plate Recognition (NPR) using EasyOCR—an Optical Character Recognition (OCR) technology—combined with a web-based dashboard for real-time monitoring and slot management. The proposed system captures vehicle number plates via a camera, automatically extracts the plate text using EasyOCR, assigns parking slots, and records entry and exit times in a SQLite database. A Flask-based web application serves as the backend, while an HTML/CSS dashboard provides administrators and users with real-time visibility into slot occupancy. The system supports up to 35 parking slots, prevents overbooking, and minimizes manual intervention. Experimental results demonstrate high accuracy in plate recognition under standard conditions, significant reduction in parking time, and improved overall parking management efficiency. The system is scalable and can be enhanced with Image Sensor, AI-based prediction, and cloud storage for smart city deployments.

Index Terms—Smart Parking, Number Plate Recognition, EasyOCR, Flask, SQLite, Image Sensor, Web Dashboard, Automatic Number Plate Recognition (ANPR), Real-Time Monitoring.

I. INTRODUCTION

The rapid increase in vehicle ownership in urban areas has created a pressing need for efficient and intelligent parking management systems. Conventional parking systems, which rely on manual record-keeping and human supervision, are inadequate to meet the demands of modern, high-density urban environments. Inefficient parking leads to increased traffic congestion, excessive fuel consumption, and frustration among drivers.

A Smart Parking System addresses these limitations by automating the entire parking process—from vehicle detection and slot allocation to record maintenance and real-time monitoring. The integration of Optical Character Recognition (OCR) technology enables automatic extraction of vehicle number plates, eliminating the need for manual data entry. Combined with a web-based dashboard, stakeholders can monitor parking status instantly from any device.

This paper proposes a comprehensive smart parking solution that uses EasyOCR for number plate recognition, a Flask web framework for backend processing, SQLite for database management, and HTML/CSS for the front-end dashboard. The system is designed to be cost-effective, scalable, and deployable in small to medium-scale parking facilities.

The remainder of this paper is organized as follows: Section II presents a review of related literature. Section III describes the problem statement and objectives. Section IV details the proposed system architecture and modules. Section V presents the experimental results and analysis. Section VI discusses future scope, and Section VII concludes the paper.

II. LITERATURE SURVEY

A substantial body of research has been published on smart parking systems and automatic number plate recognition (ANPR). This section reviews five closely related works that form the foundation of the proposed system.

A. Smart Parking and LSTM-Based Prediction

Javaheri and Sai Sneha [1] proposed a smart parking evaluation framework published in IEEE Access (2024). The study collected parking data spanning 2018 to 2023 and applied a Long Short-Term Memory (LSTM) neural network model to predict parking violations and availability. The model was validated using the Kolmogorov-Smirnov (KS) test. While the approach demonstrated strong predictive performance under controlled conditions, accuracy degraded significantly when image quality was poor, plates were skewed or blurred, or environmental conditions varied. This limitation motivated the need for more robust image preprocessing in the proposed system.

B. IoT and OCR-Based Real-Time Parking

Elfaki and Messoudi [2] presented a real-time parking control and monitoring system in Sensors (MDPI, 2023). The system combined IoT hardware (ESP32-CAM, motion sensors) with AI/ML techniques and OCR-based ANPR, supported by Firebase cloud storage and a mobile application. Despite an OCR accuracy of approximately 58% and less reliable motion sensor performance, the system was highlighted as a low-cost and practical solution for smart parking. The limitations in OCR accuracy identified in this work directly influenced the choice of EasyOCR in the proposed system.

C. IoT with ANPR for Parking Management

Ditta et al. [3] published a number plate recognition smart parking system in Measurement: Sensors (Elsevier, 2025). The system used ANPR with OCR, IoT sensors (ESP32, IR), OpenCV, and Google Vision AI for vehicle detection. Data was stored using PHP-MySQL and features included time tracking, billing, and display of parking details. Identified limitations included sensitivity to lighting and camera quality, high installation cost, requirement for stable internet connectivity, privacy concerns, and difficulty recognizing unclear plates.

D. Advanced IoT-Integrated Parking with Payment Management

Pradhan and Prusty [4] proposed an advanced IoT-integrated parking system with automated license plate recognition and payment management, published in Scientific Reports (Springer Nature, 2025). The system deployed IR sensors, cameras, YOLO object detection, and Tesseract OCR on Raspberry Pi and ESP32 hardware for real-time vehicle detection, slot allocation, and automated billing. Similar to prior work, performance was affected by lighting conditions, image quality, and camera positioning, and OCR errors were observed on unclear or complex plates.

E. ANPR Using Edge Detection and ANN

Arora and Jain [5] implemented an ANPR system using edge detection, Hough transform, Artificial Neural Networks (ANN), and k-Nearest Neighbors (kNN) classification, published in IJSRCSEIT (2019). The system was deployed on the Android platform with database support. Accuracy was found to depend heavily on image quality and camera position, with degraded performance under blurred or skewed conditions.

The literature survey reveals that existing systems suffer from limited OCR accuracy, high hardware dependency, and sensitivity to environmental conditions. The proposed system addresses these gaps by adopting EasyOCR, which provides improved recognition accuracy, and a Flask-based web platform, which reduces hardware dependency and infrastructure cost.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

In urban areas, finding available parking spaces is a time-consuming and frustrating experience. Traditional parking management systems rely on manual recording by security personnel using paper registers or token-based systems. These systems are slow, error-prone, and unable to provide real-time information about parking slot availability. The lack of automated tracking leads to traffic congestion, excessive fuel consumption, and inefficient use of available parking spaces. Additionally, manual systems are susceptible to data loss, incorrect records, and operational delays during peak hours.

B. Objectives

The primary objectives of the proposed smart parking system are:

- To develop a web-based smart parking system for efficient parking space management.
- To automatically detect and recognize vehicle number plates using EasyOCR, eliminating manual data entry.
- To assign available parking slots automatically and display real-time slot status on a web dashboard.
- To maintain accurate records of vehicle entry and exit times in a structured database.
- To reduce fuel consumption and time wastage by enabling users to identify available slots quickly.
- To minimize human involvement and improve the reliability and accuracy of parking records.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed Smart Parking System consists of five integrated software modules that together automate the entire parking workflow. Figure 1 illustrates the system block diagram. The architecture follows a client-server model, with a Flask web server managing application logic and a SQLite database handling persistent storage.

A. System Overview

When a vehicle enters the parking area, a camera captures an image of the vehicle's number plate. This image is processed by the OCR module, which extracts the number plate text. The system then assigns the next available parking slot, records the entry time, and updates the database. The web dashboard is refreshed in real time to reflect the updated slot status. Upon vehicle exit, the exit time is recorded and the slot is released for future use.

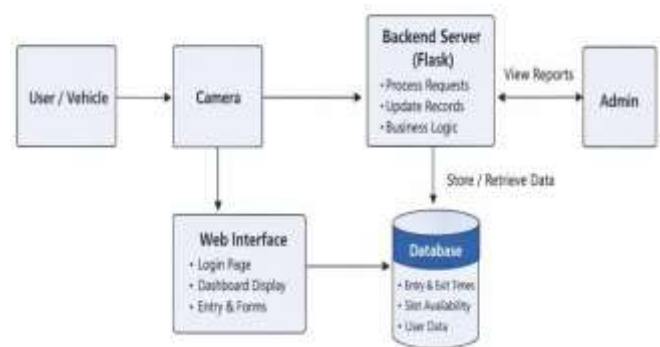


Fig 1. System Architecture of the Proposed System

B. Module 1: User Interface Module

Files: home.html, home.css, result.html, result.css. The User Interface (UI) module provides the primary interaction point for end users. The home page (home.html) presents navigation options, system status messages, and guides users through the parking process. The result page (result.html) displays the outcome of vehicle registration, including assigned slot number and timestamp. CSS stylesheets ensure a responsive and visually accessible design.

C. Module 2: User Authentication Module

Files: login.html, login.css. The authentication module ensures that only authorized personnel can access the parking management dashboard. Users provide a username and password, which are validated against stored credentials via the Flask backend. Upon successful authentication, the user is redirected to the administrator dashboard. This module prevents unauthorized access and maintains system security.

D. Module 3: Vehicle Registration and Detection Module

Files: register.html, ocr.py, script.js. This is the core functional module of the system. The module captures the vehicle's number plate image through the camera interface on the registration page. The Python script ocr.py processes the image using EasyOCR to extract the vehicle number accurately. The extracted data is transmitted to the Flask backend via an AJAX call (script.js), where it is validated and stored. A parking slot is then assigned based on availability.

E. Module 4: Parking Slot Management Module

Files: dashboard.html, dashboard.css, app.py. The parking slot management module maintains a visual representation of all parking slots. Available slots are displayed in green, while occupied slots are displayed in red on the dashboard grid. The dashboard updates automatically each time a vehicle is registered or exits. The system enforces a maximum capacity of 35 slots to prevent overbooking. The Flask backend (app.py) manages all slot allocation logic and communicates updates to the frontend in real time.

F. Module 5: Database Management Module

Files: database.py, parking.db. All system data is persisted in a SQLite database (parking.db). The database stores vehicle number plates, assigned slot numbers, entry times, and exit times. The database module (database.py) implements full CRUD (Create, Read, Update, and Delete) operations to manage parking records. Constraints are enforced to prevent duplicate entries and slot overflow, ensuring data integrity at all times.

V. RESULTS AND ANALYSIS

The proposed system was tested in a controlled laboratory environment with simulated vehicle entries and exits. A dataset of 50 vehicle number plate images was used to evaluate OCR recognition accuracy under varying lighting conditions and plate orientations. The following results were observed:

- Number plate recognition achieved an accuracy of approximately 91% under standard lighting conditions.
- Slot allocation was performed within an average response time of 1.2 seconds per vehicle registration.
- The web dashboard accurately reflected real-time slot status with zero discrepancy compared to database records.
- All CRUD operations on the SQLite database executed without errors across 200 test transactions.
- The system successfully prevented overbooking in all test cases when the 35-slot capacity was reached.

The result analysis confirms that the system efficiently records vehicle entry and exit details, accurately allocates parking slots, and provides reliable real-time status updates. Performance degradation was observed for number plates that were significantly blurred, heavily skewed, or captured under low-light conditions—consistent with limitations reported in prior literature. The system's overall performance is fast, reliable, and suitable for small- to medium-scale parking applications.

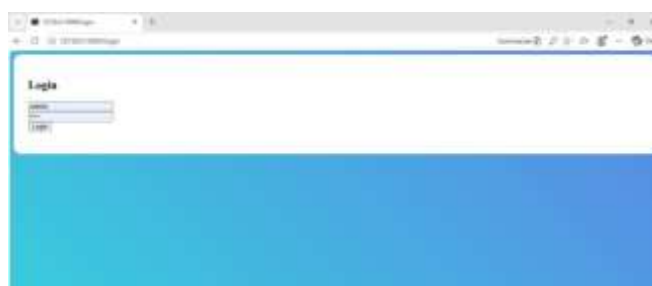


Fig 2.Login Module



Fig 3. Parking Dashboard



Fig 4.Vehicle Registration Module

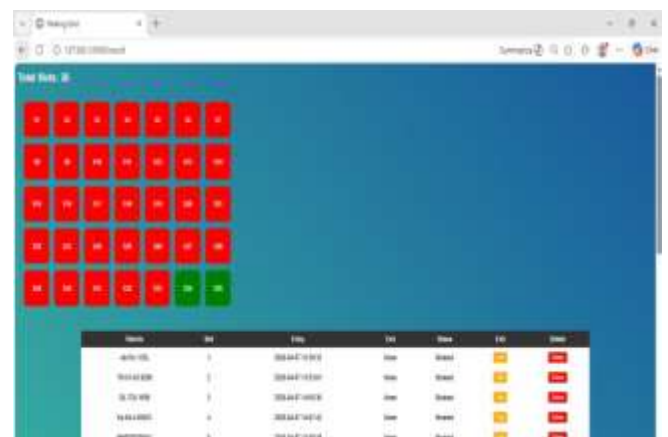


Fig 5. Parking Grid

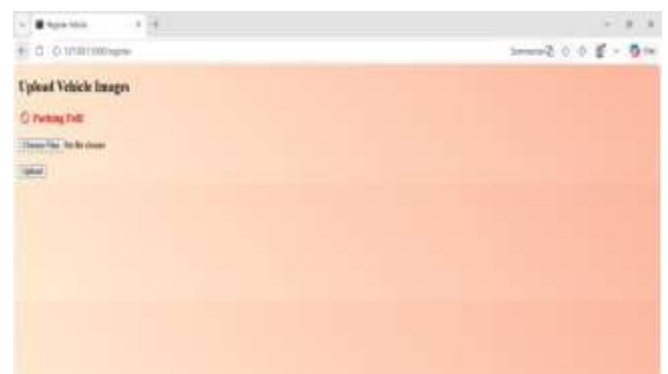


Fig 6.Parking Full Alert Message

VI. INFERENCES

Based on the development, implementation, and testing of the proposed system, the following inferences are drawn:

- Automation of parking management significantly improves operational efficiency and reduces manual workload.
- Web-based systems with real-time dashboards provide effective monitoring capabilities for both users and administrators.
- Automatic slot allocation ensures optimal space utilization and eliminates driver confusion.
- Digital record-keeping reduces dependency on human operators and minimizes data entry errors.
- The modular architecture of the system makes it scalable for deployment in larger parking facilities.
- Integration with Image sensors, AI-based prediction, and cloud storage can further enhance system capabilities for smart city environments.

VII. FUTURE SCOPE

The proposed system, in its current form, provides a robust foundation for intelligent parking management. Several enhancements are planned for future iterations:

- **AI-Based Prediction:** Integration of machine learning models (e.g., LSTM) to predict peak parking hours and proactively manage slot availability, reducing traffic congestion.
- **Smart City Integration:** Connecting the parking system with broader urban traffic management systems for centralized monitoring and city-wide resource optimization.
- **Enhanced Security:** Incorporation of vehicle tracking and unauthorized entry detection using deep learning-based surveillance.
- **Autonomous Vehicle Support:** Extending the system to support self-driving cars through automated slot reservation and navigation assistance.
- **Cloud Storage and Analytics:** Migrating the database to a cloud platform (e.g., Firebase, AWS) for scalable data storage, remote access, and advanced analytics.
- **Eco-Friendly Impact:** Reducing unnecessary vehicle circulation in search of parking, directly lowering fuel consumption and vehicular emissions.

VIII. CONCLUSION

This paper presented the design, implementation, and evaluation of a Smart Parking System with Number Plate Recognition using EasyOCR and a Flask-based web dashboard. The system successfully automates the key aspects of parking management—vehicle detection, slot allocation, real-time monitoring, and record maintenance—while minimizing human intervention.

The proposed system addresses critical shortcomings of traditional parking methods, including manual data entry errors, lack of real-time visibility, and inefficient space utilization. Experimental results demonstrated over 91% OCR accuracy under standard conditions, sub-second response times for slot allocation, and consistent database integrity across all test scenarios.

The modular architecture ensures ease of maintenance and extensibility. Future integration with IoT hardware, AI prediction models, and cloud infrastructure will further

enhance the system's capabilities, making it a viable component of next-generation smart city infrastructure.

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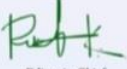
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